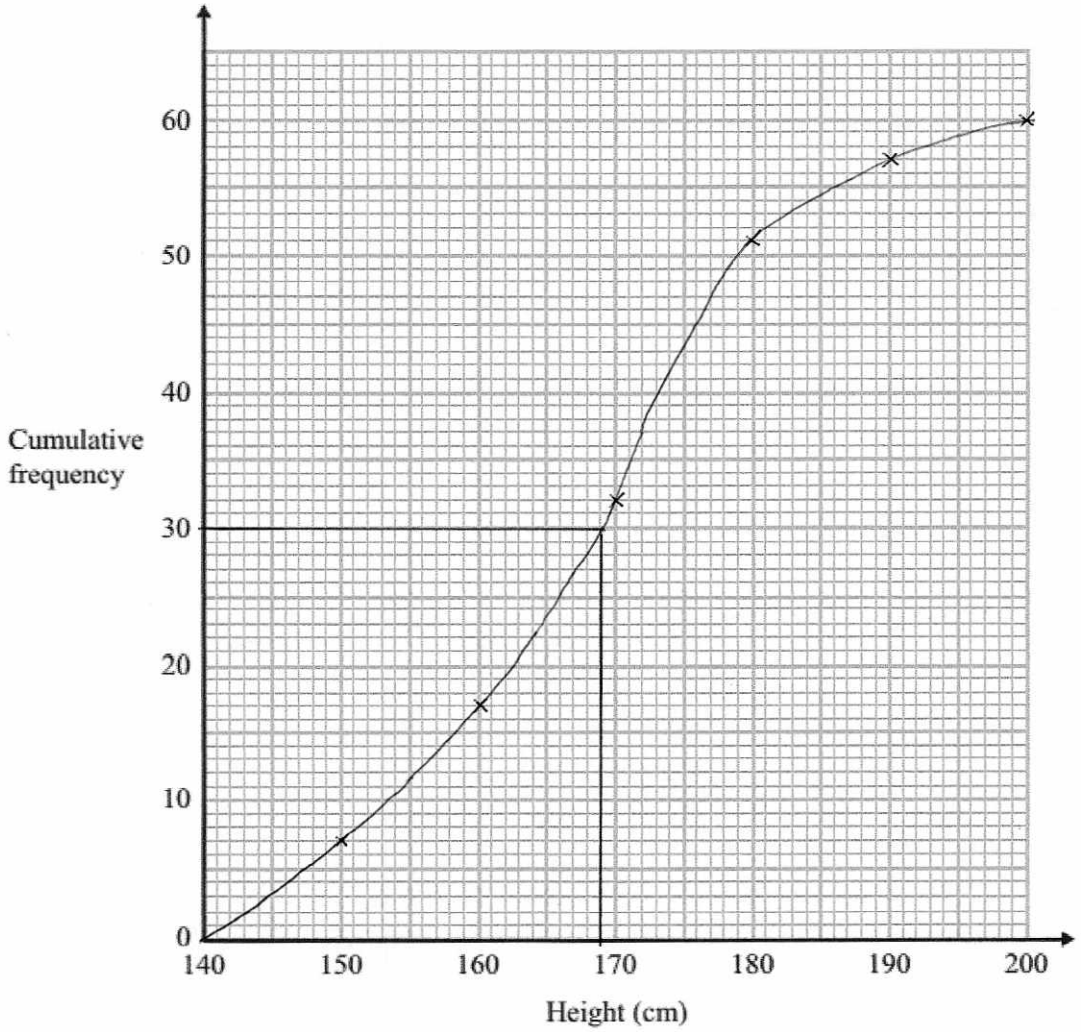
1 The cumulative frequency table shows the height, in cm, of some tomato plants.

Height	Cumulative Frequency
$140 < h \leqslant 150$	7
$140 < h \leqslant 160$	17
$140 < h \leqslant 170$	32
$140 < h \leqslant 180$	51
$140 < h \leqslant 190$	57
$140 < h \leqslant 200$	60

(a) On the grid, plot a cumulative frequency graph for this information.



(2)

(b) Find the median height.

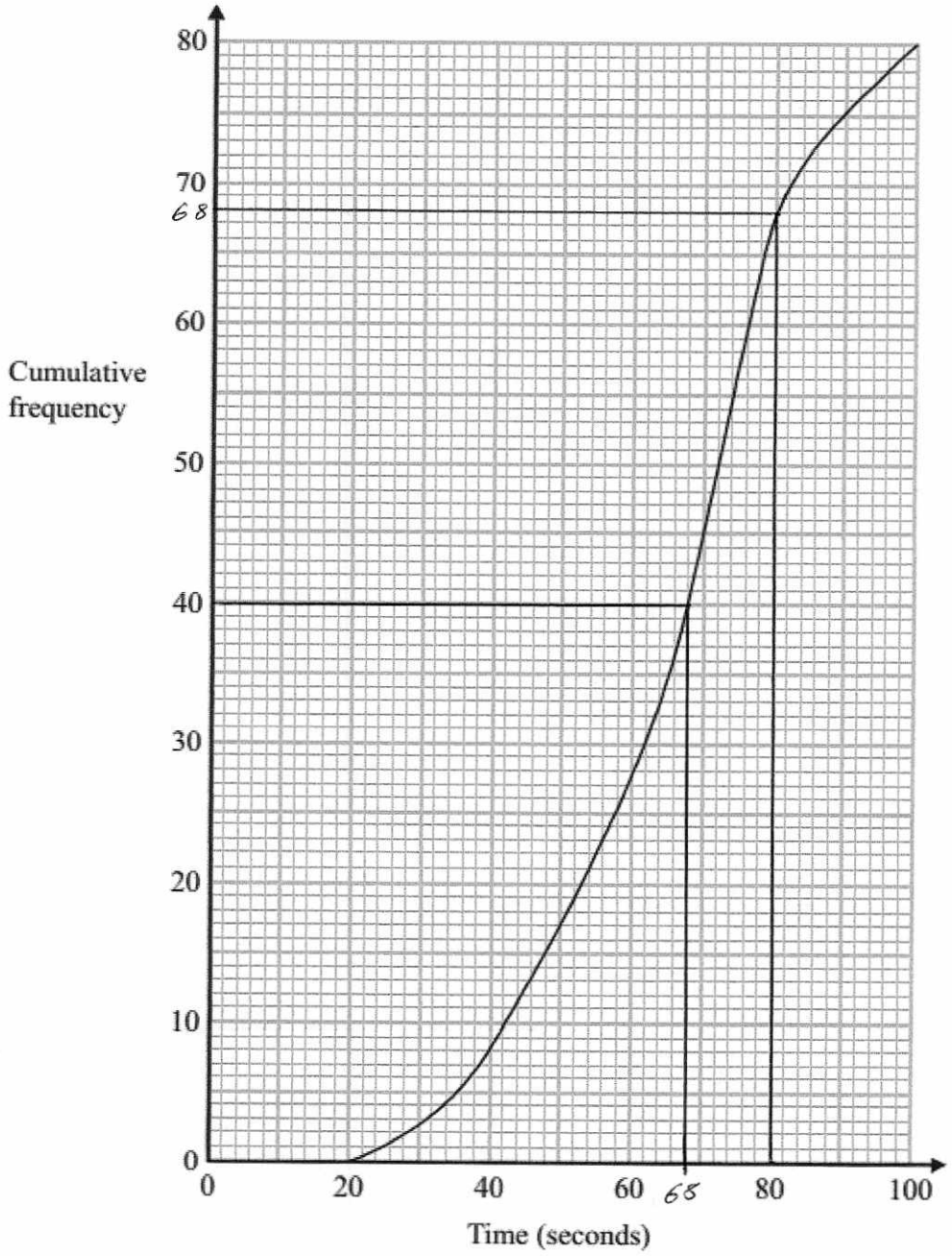
an estimate for e

.....169.....cm

168 – 169 (1)

(Total for question 1 is 3 marks)

2 The cumulative frequency graph gives some information the times it took people to complete a challenge.



(a) Find the median time.

68 seconds

(1)

(b) Find the number of people who took longer then 80 seconds to complete the challenge.

80 − 68 = 12

12

(1)

(Total for question 2 is 2 marks)

3 The frequency table shows the weight, in kg, of some cats.

Weight (kg)	Frequency
$0 < w \leqslant 1$	8
$1 < w \leqslant 2$	10
$2 < w \leqslant 3$	21
$3 < w \leqslant 4$	19
$4 < w \leqslant 5$	13
$5 < w \leqslant 6$	9

(b) Complete the cumulative frequency table

(1)

Weight (kg)	Cumulative Frequency
$0 < w \leqslant 1$	8
$0 < w \leqslant 2$	18
$0 < w \leqslant 3$	39
$0 < w \leqslant 4$	58
$0 < w \leqslant 5$	71
$0 < w \leqslant 6$	80

(b) On the grid opposite draw a cumulative frequency graph for this information.

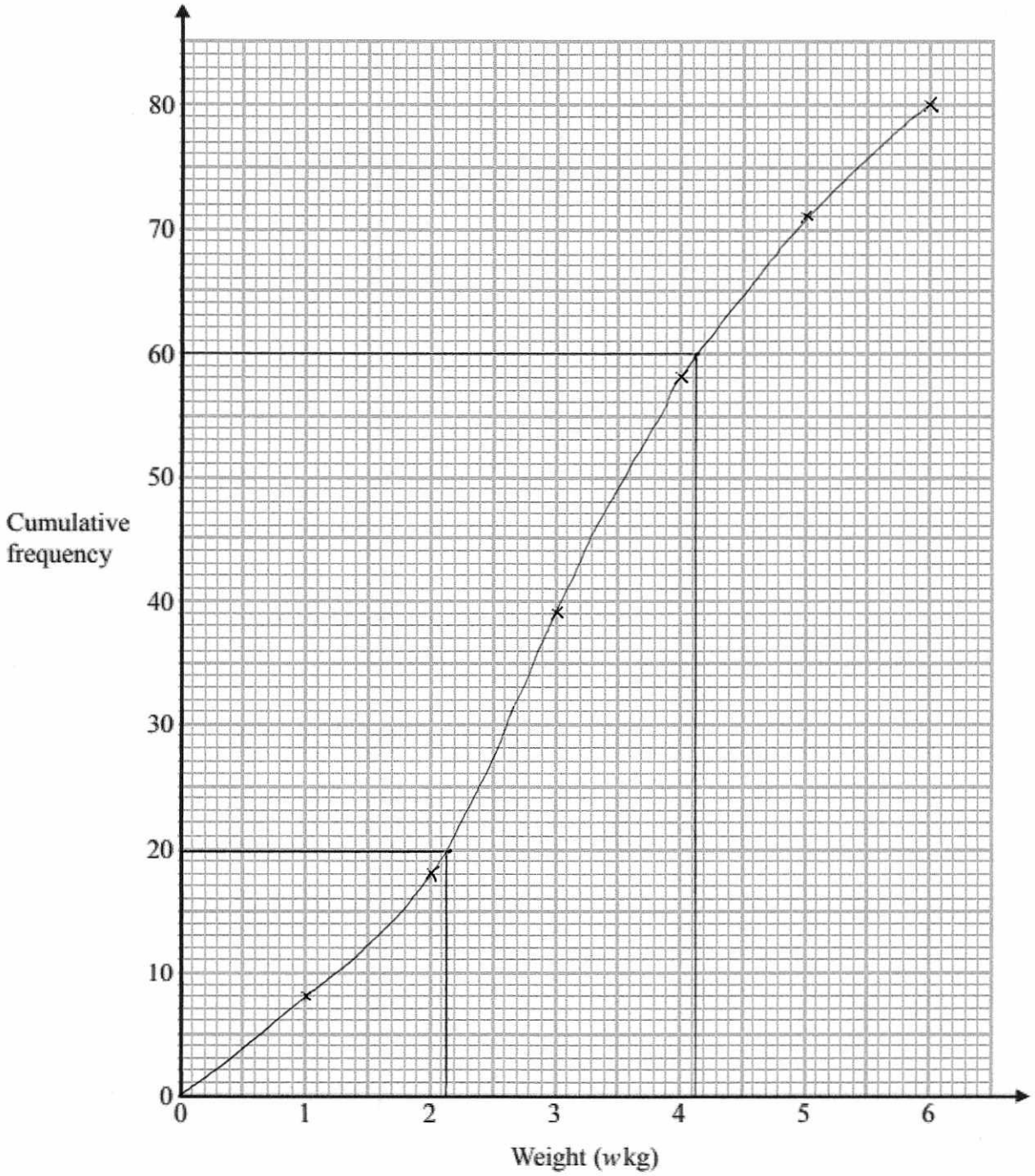
(2)

(c) Use your cumulative frequency graph to find an estimate for the interquartile range.

4.1 − 2.1 = 2

.....2.....kg

1.9 − 2.1 **(2)**

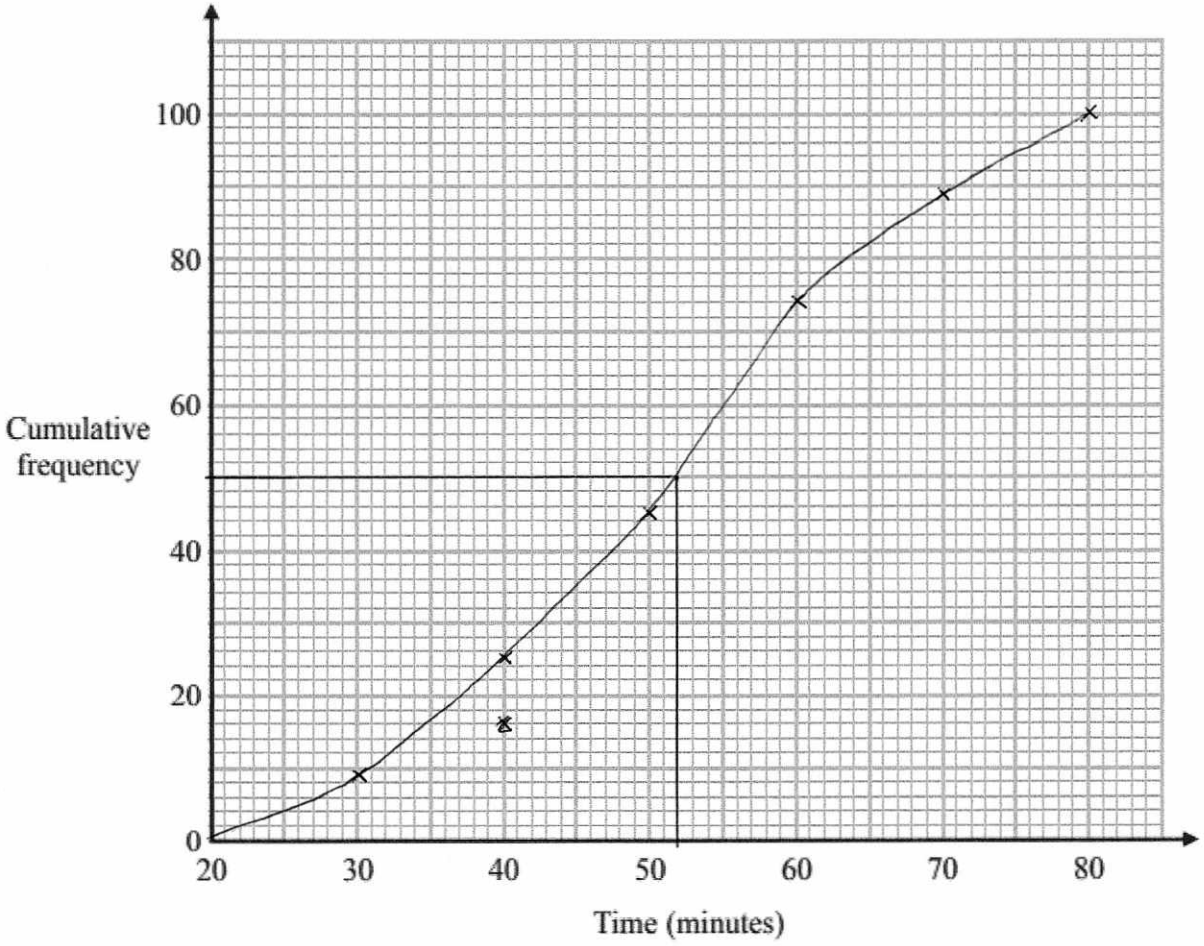


(Total for question 3 is 5 marks)

4 The frequency table shows the time taken for 100 people to travel to an event.

Time (minutes)	Frequency	C.F
$20 < t \leqslant 30$	9	9
$30 < t \leqslant 40$	16	25
$40 < t \leqslant 50$	20	45
$50 < t \leqslant 60$	29	74
$60 < t \leqslant 70$	15	89
$70 < t \leqslant 80$	11	100

(a) On the grid, plot a cumulative frequency graph for this information.



(2)

(b) Find an estimate for the median time taken.

.....52.....minutes

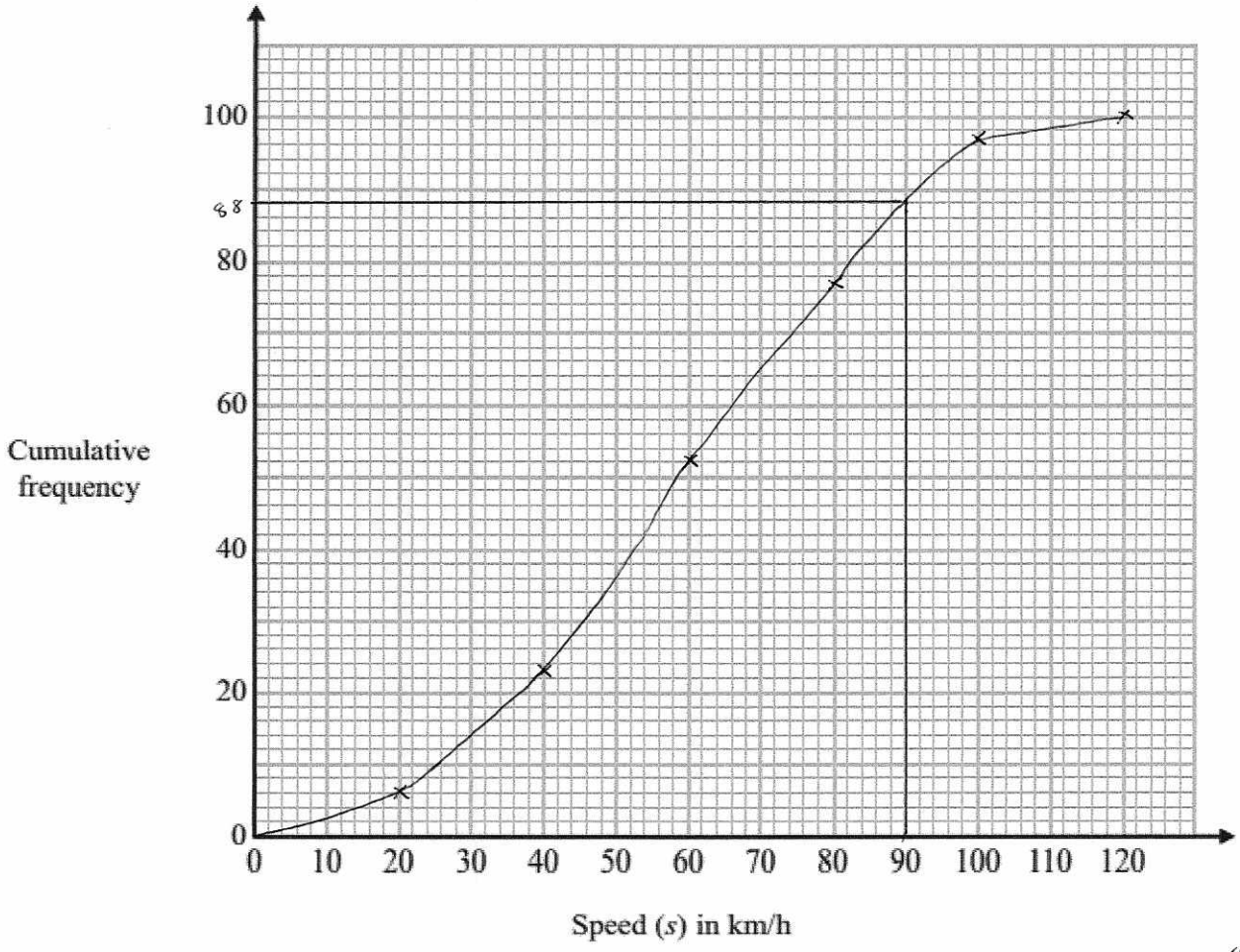
51 - 53 (1)

(Total for question 4 is 3 marks)

5 The frequency table shows the speeds of 100 cars.

Speed (km/h)	Frequency	C.F
$0 < s \leqslant 20$	6	6
$20 < s \leqslant 40$	17	23
$40 < s \leqslant 60$	29	52
$60 < s \leqslant 80$	25	77
$80 < s \leqslant 100$	20	97
$100 < s \leqslant 120$	3	100

(a) On the grid, plot a cumulative frequency graph for this information.



(b) Find an estimate for the number of cars travelling over 90 km/h. (2)

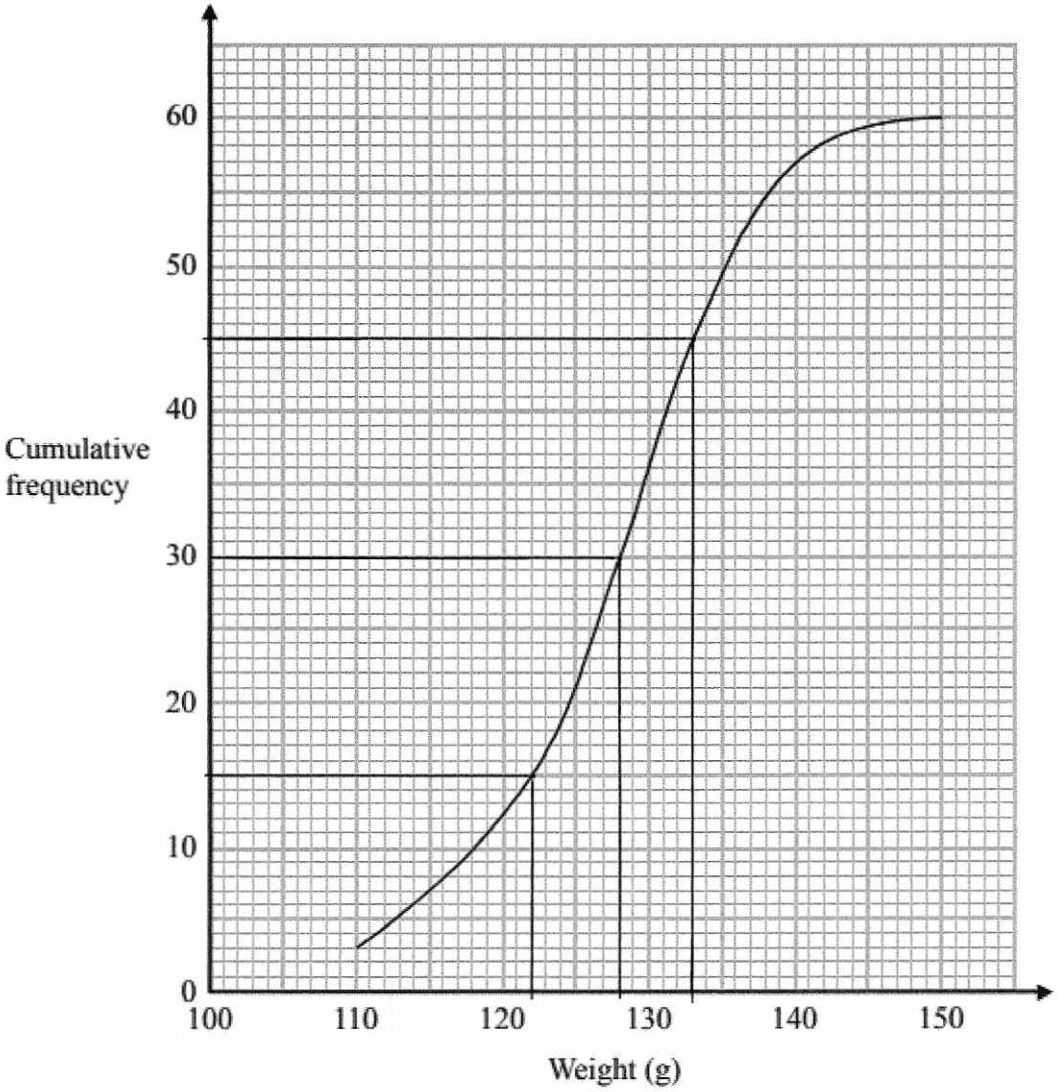
100 − 88 = 12

12

10 − 14 (1)

(Total for question 5 is 3 marks)

6 The cumulative frequency graph gives some information about the weights of some objects.



(a) Find the median weight.

128 g

(1)

(b) Find the inter quartile range.

133 − 122 = 11

11 g

(2)

(Total for question 6 is 3 marks)